Dam AI Guardian

Predict the Unseen. Prevent the Inevitable.

Revolutionising dam safety through artificial intelligence and real-time monitoring for a safer tomorrow.

HackData - SNU | 🚀 Team Technovates

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The Crisis No One Sees Coming

India's aging dam infrastructure poses a silent threat that demands immediate technological intervention.

Hidden Structural Threats

Undetected seepage and micro-cracks silently compromise building foundations and infrastructure stability

Aging Infrastructure

Over 70% of India's 5,000+ dams are past their 25-year operational threshold, requiring vigilant monitoring

Manual Detection Failure

Current manual inspection methods are slow, reactive, and lack predictive capabilities for early warnings

Human Cost

Thousands of lives lost annually due to delayed detection and insufficient early warning systems

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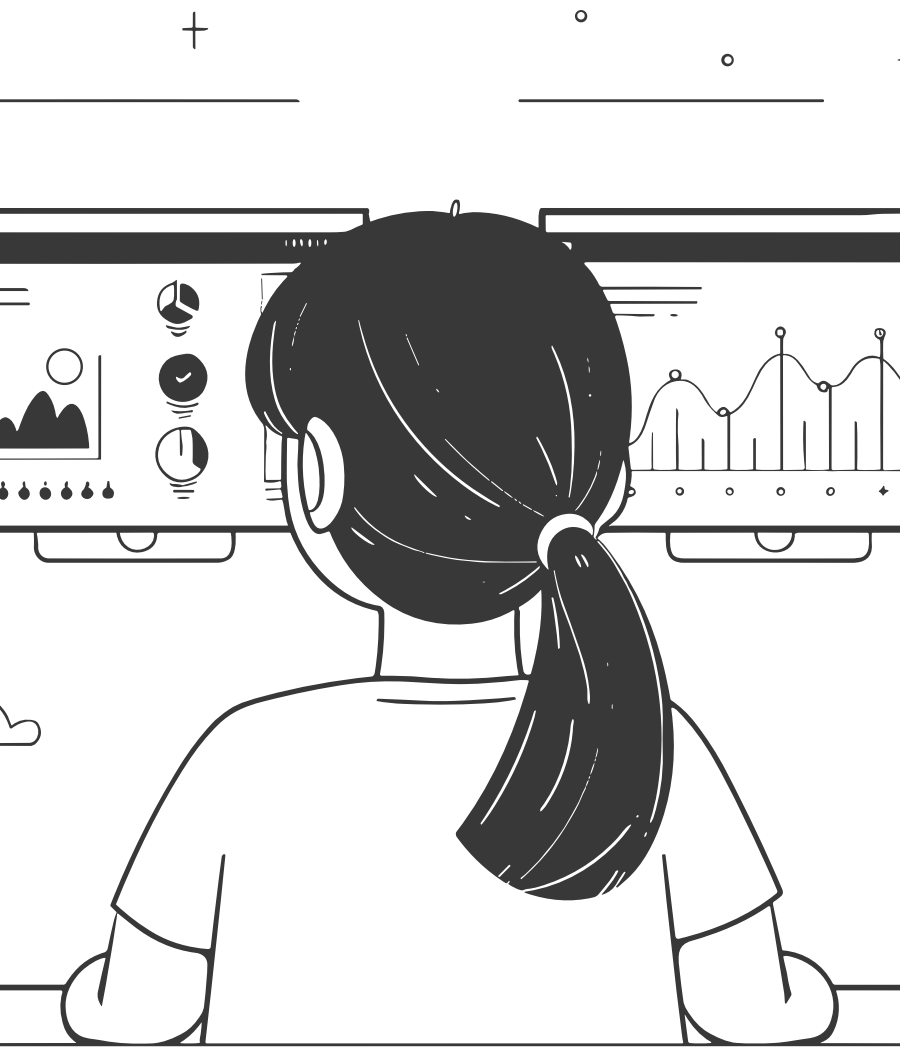

One AI System. Total Protection.

01

Real-Time Monitoring

IoT sensors continuously collect structural health data whilst live weather feeds provide environmental context for comprehensive risk assessment

02

AI Risk Engine

Machine learning models classify risk levels as Low, Medium, or High instantly using multi-dimensional data analysis and pattern recognition

03

Instant Alerts

Multi-channel notifications via SMS, WhatsApp, and dashboard alerts delivered in both English and Hindi for maximum accessibility

Additional Features: PWA mobile app • Voice-activated controls • Multi-language support • Real-time visualisation

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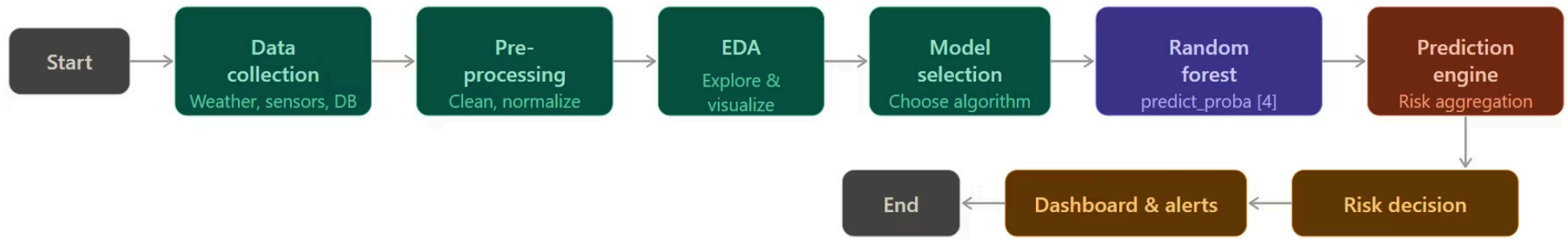
How It Works

Our system transforms raw sensor data into life-saving predictions through an intelligent, multi-stage processing pipeline.



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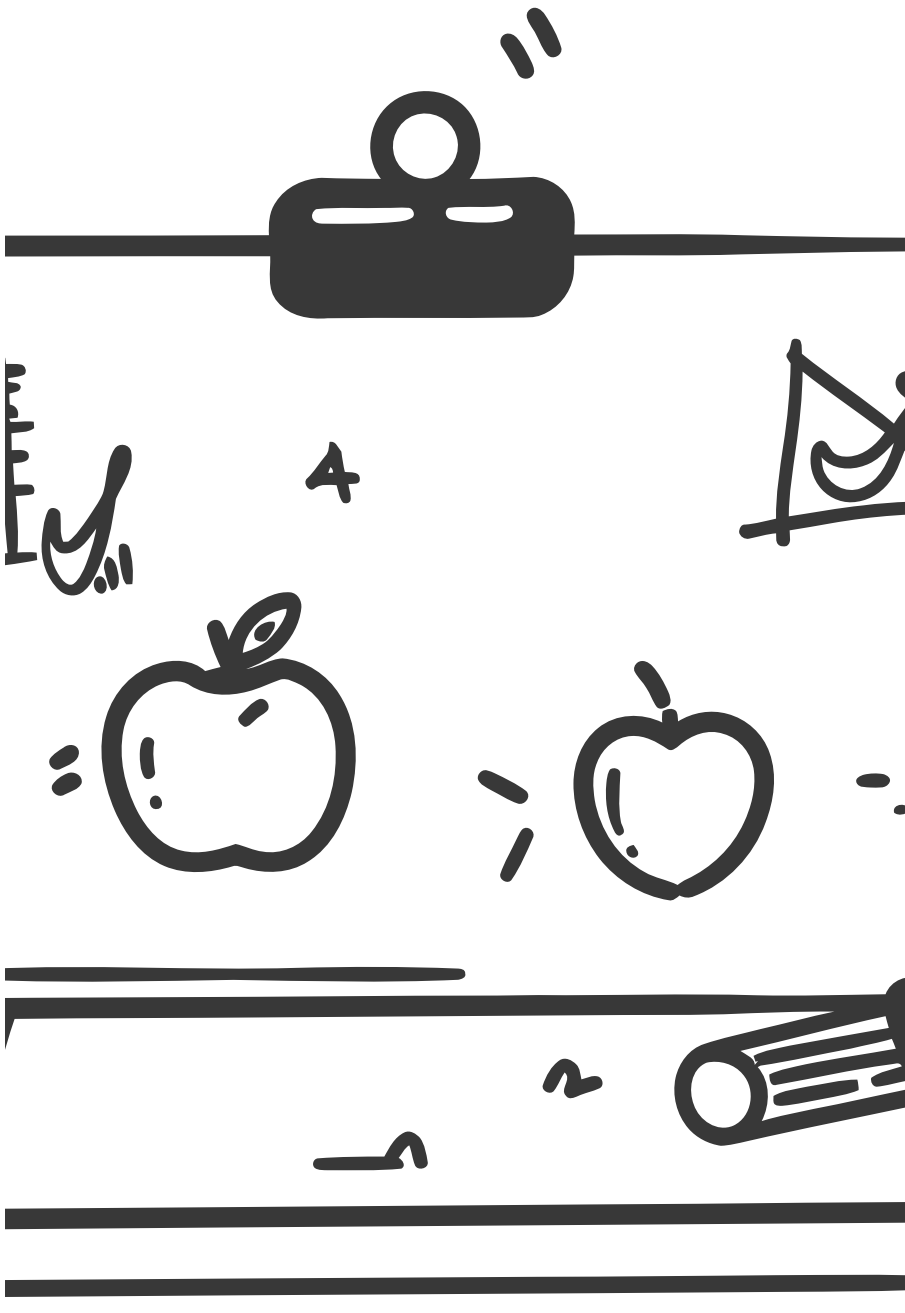
Powered by Real Data & Smart Models

Input Data Sources

- **Dam Infrastructure Data:** Location coordinates, construction age, storage capacity, crack dimensions, maintenance history, and structural specifications
- **Environmental Parameters:** Real-time rainfall intensity, humidity levels, ambient temperature, soil composition, and groundwater levels
- **Weather Integration:** Live weather forecasts and historical patterns via Open-Meteo API for predictive modelling

Machine Learning Models

- **Logistic Regression:** Binary classification for risk probability estimation
- **Random Forest:** Ensemble method for robust predictions across diverse conditions
- **Gradient Boosting:** Sequential error correction for enhanced accuracy



Low Risk

Normal operations continue

Medium Risk

Enhanced monitoring required

High Risk

Immediate action needed

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Built for the Real World

Real-Time Visualisation

Interactive dashboards display sensor data with live updates and historical trend analysis for comprehensive monitoring

Weather Integration

Live weather data from Open-Meteo API provides contextual environmental factors for accurate risk assessment

Twilio Alerts

Automated SMS and WhatsApp notifications via Twilio API and WhatsApp Green API for instant stakeholder communication

Multi-Language Support

English and Hindi interface ensures accessibility across diverse user groups and regional stakeholders

PWA Mobile App

Progressive web app installs on any device without app store downloads, providing offline functionality and push notifications

Voice Commands

Voice-activated alert systems allow hands-free operation for field personnel and emergency response teams

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The Technology Behind the Guardian

Frontend Architecture

React 18 • TypeScript • Tailwind CSS • Vite for lightning-fast rendering and responsive design across all devices

Backend Infrastructure

Node.js • Express.js • Socket.IO for real-time data streaming and bidirectional communication channels

ML & AI Engine

Python • Scikit-learn • Flask API for machine learning model training, evaluation, and prediction serving

External Services

Twilio • WhatsApp Green API • Open-Meteo • MongoDB • Vercel for notifications, weather data, storage, and deployment

System Flow: React/Vite frontend → Node/Express/Socket.IO server → Flask API → ML Models → Dashboard & Alert System

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Why This Matters & How We Scale

Market Opportunity

$11B+ global dam safety market growing at **15% CAGR**

5,000+ dams in India with 70% requiring modern monitoring systems

Target Customers

NDMA & Government Bodies for national infrastructure protection

Smart City Authorities for urban development projects

Construction & Real Estate firms for building safety compliance

Insurance Companies for risk assessment and premium calculation

Revenue Streams

SaaS Subscriptions: Tiered plans for different organisation sizes

Pay-Per-Use API: Integration services for third-party platforms

Government Tenders: Large-scale infrastructure monitoring contracts

Premium Features: Computer vision crack analysis add-on

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